Report of AG News Topic Classifier

May 10, 2026

1 Introduction

This project aims to address the problem of news topic classification in the context of information retrieval. Its objective is to classify short news texts into one of the following four categories: World, Sports, Business, and Sci/Tech. Such classification helps organize document collections, filter search results, and present retrieved information by topic, thereby supporting information retrieval systems.

The challenge of this project is that news articles on the same topic may use different vocabulary. For example, technology news may mention software, space research, artificial intelligence, or hardware, but these topics often do not share many keywords in common. To address this issue, the project employs sentence embedding to represent text as dense semantic vectors, enabling the classifier to make judgments based on the overall semantic meaning of the text rather than relying on superficial word matching.

2 Dataset

The dataset used in this project is a processed subset of the AG News Classification Dataset from Kaggle. The original dataset contained three fields: Class Index, Title, and Description. For this assignment, I created a processed version by combining the title and description into a single text field, mapping numeric labels to readable class names, and creating a balanced training and test subset.

The final processed dataset contains 4,000 training examples and 1,000 test examples, and was uploaded to Hugging Face.

3 Method

The classification pipeline consists of two main steps. First, each news text is encoded into a sentence embedding using sentence-transformers/all-MiniLM-L6-v2. Second, a Logistic Regression classifier is trained on these embeddings to predict the topic label.

This approach was chosen because it is simple, efficient, and directly addresses the assignment requirement of training a classifier using embeddings. The trained model was saved as a joblib file and uploaded to Hugging Face.

A Gradio demo was also created and deployed on Hugging Face Spaces.

4 Results

The model was evaluated on a balanced test set of 1,000 examples. The results are shown in Table 1.

Class	Precision	Recall	F1-score	Support
World	0.91	0.85	0.88	250
Sports	0.96	0.95	0.96	250
Business	0.82	0.87	0.84	250
Sci/Tech	0.86	0.86	0.86	250
Accuracy	–	–	0.8830	1000
Macro average	0.88	0.88	0.8834	1000

Table 1: Evaluation results on the balanced test set.

The model achieved an overall accuracy of 0.8830 and a macro F1-score of 0.8834. Sports was the easiest category to classify, while Business was the most difficult, possibly because business news can overlap with world events and technology-related topics.

5 Working with AI Tools

During the implementation process, I used ChatGPT and Claude to assist with development, primarily to structure Python scripts, understand parts of the code logic, complete the deployment processes on Hugging Face and GitHub, and modify the LaTeX format. However, AI-generated code still requires manual review and modification, such as verifying data paths, label mappings, model repository names, and the loading logic for the Gradio demo. While using AI tools certainly accelerated the development process, it also required me to understand the purpose of each code block. I needed to figure out how sentence embeddings are generated, how Logistic Regression uses these embeddings as input features for classification. Therefore, this project was not only a practical exercise in using AI tools to complete a programming task, but it also allowed me to practice critically reviewing and refining AI-generated output.

6 Links

- GitHub repository: https://github.com/Thisisruiii/ir-embedding-news-classifier
- Hugging Face dataset: https://huggingface.co/datasets/Thisisruiii/ag-news-topic-classification-processec
- Hugging Face model: https://huggingface.co/Thisisruiii/AG-News-Classifier
- Hugging Face demo: https://huggingface.co/spaces/Thisisruiii/AG-News-demo